MAIA

Monitoring Assistant & Intelligent Automation

Intelligent Monitoring & Auto-Healing Platform

Operator Onboarding Guide

Version 2.2 · July 2026

1 Welcome to MAIA

MAIA (Monitoring Assistant & Intelligent Automation) is an intelligent automation platform that monitors your data pipelines, detects failures, classifies them, and either fixes them automatically or guides you through resolving them. This guide walks you through everything you need to know as an operator.

What MAIA Does for You

- Watches your pipelines 24/7 across 4 source types (log files, databases, APIs, XML files)
- Detects failures automatically using configurable scan rules
- Classifies each failure by error type using pattern-matching rules
- Suggests fixes from a policy catalogue — including multi-step composite fixes
- Auto-heals known issues without your intervention (when configured)
- Logs every action immutably for audit compliance

2 Logging In

MAIA uses local username/password authentication with three role tiers.

Step	Action
1. Open the MAIA URL	Your admin will provide the URL (e.g. https://server/Maia.Client/).
2. Enter credentials	Your admin creates your account. Default admin account: admin/admin (must be changed on first login).
3. Change password	If this is your first login or your password was reset, you **must** change it before accessing any screen. This cannot be skipped in production.
4. You're in	You'll see the Dashboard. Your role determines what you can access.

Your Role	What You Can Do
User	View dashboard, failures, scan history. Read-only — cannot approve or configure.
Operator	Everything User can do + approve/reject/retry fixes, mark resolved, trigger scans, set auto-heal.
Administrator	Everything Operator can do + configure jobs/rules/policies, manage users, view audit log, control worker.

3 The Dashboard

The Dashboard is your home screen. It gives you an instant overview of the system's health.

KPI Tiles (top row):

Tile	What It Means
Active Failures	Failures currently in Failed status. Click to drill down.
Awaiting Action	Failures waiting for your review (have recommendations but no decision yet).
Manual Required	Failures that need manual intervention — auto-heal couldn't resolve them.
Fix Failures Today	Failures where an automated fix failed today. Red pulse = needs attention.
Unconfigured	Failures with no classification or fix policy. Admin needs to add rules.
Resolved Today	How many were resolved today (auto + manual breakdown). De-emphasized — informational only.

Analytics Charts:

Below the KPIs: **Errors Over Time** (stacked area chart, 24h/7d/30d range toggle) shows failure trends by error type. **Failures by Job** (horizontal bar) shows which jobs fail most. Both update every 5 seconds.

Bottom panels:

Recent Failures shows the latest 10 failures — click any to open the failure drawer. **Monitored Jobs** shows all active jobs with compact status rows (click to expand details).

Dashboard Tips

- Everything auto-refreshes every 5 seconds — no manual refresh needed
- If the top-bar pill turns amber with '■ Reconnecting...', the server connection is down — data shown is stale
- Click any KPI tile to drill down to the matching failures list filter
- Dark mode available: click the sun/moon icon in the top bar

4 Reviewing Failures

The Failures screen (/failures) is where you spend most of your time. It shows all detected failures with filters and sorting.

Finding failures:

Control	How It Works
Status filter	Failed, Resolved, ManualRequired, AwaitingManualAction — or All
View filter	Active (default), Resolved Today, Manual Required, Fix Failed Today
Search	Free-text search across job name, step name, error message. Debounced 250ms.
Sort	Click any column header (Job, Error Type, Detected, Status) to sort. Click again to flip direction.
Pagination	Server-side. ↓ on last row auto-loads next page; ↑ on first row loads previous.

The Failure Drawer:

Click any failure row to open a 760px slide-in drawer on the right. The drawer shows:

Section	What You See
Error Context	Job name, error type, step name, raw error message, source file path, detected timestamp.
Stage Pipeline	Visual progress: Detected → Classified → Recommended → [Acknowledged or Manual] → Fixed
Recommendations	Each recommendation shows: suggested action, confidence score, auto-heal status, fix category. Composite fixes show a bulleted step list.
Execution History	Collapsible per-attempt list: ✓/✗ per step, newest attempt first. Red banner counts failures in the latest attempt only.
Action Buttons	Approve (runs fix now), Acknowledge (for manual fixes), Reject, Retry (re-arms a failed fix), Mark Resolved (confirms off-system work done)

Keyboard Shortcuts in Failures

- ↑/↓ — Navigate between failure rows (auto-loads next/prev page at boundaries)
- Enter — Open drawer for the focused row
- Esc — Close the drawer
- Ctrl/Cmd+K — Open command palette (jump to any screen, search failures by ID, find jobs)

5 Taking Action on Failures

When you review a failure with recommendations, you have several options:

Action	What Happens
Approve	Runs the recommended fix immediately. For automated fixes (API/SP/Script/CopyFile), the fix executes on the same request. Status → Resolved (success) or ManualRequired (failure). For composite fixes, all steps run in order (best-effort).
Acknowledge	For Manual-category recommendations only. Signals that you've seen it and will handle it offline. Status → AwaitingManualAction. The [Approve] button reads [Acknowledge] for these.
Reject	Declines the recommendation. If it was the last pending recommendation on the failure, status → ManualRequired. Otherwise the failure stays at Failed for other recommendations to be acted on.
Retry	Available when a previous fix execution failed. Re-arms the recommendation with the CURRENT policy (in case the admin fixed the policy) and runs it again. Status resets to Failed → re-executes.
Mark Resolved	Available when status is AwaitingManualAction or ManualRequired. Confirms you've completed the off-system work. Status → Resolved.

Note: if a background auto-heal resolves a failure while you have the drawer open, the action buttons gracefully disable with an 'Already executed' note — they won't vanish under your cursor.

6 Auto-Heal

Auto-heal lets MAIA resolve known, recurring issues automatically without your intervention. It runs on every worker tick (every 5 seconds when jobs are claimed).

How to enable auto-heal:

Step	Action
1. Go to Recommendations	Navigate to the Recommendations screen.
2. Find the error type	Filter or search for the recommendation matching the error type you want to auto-heal.
3. Toggle Auto-Heal	Click the auto-heal toggle. This mutates the FixPolicyRule.IsAutoHealEligible flag for that (JobType, ErrorType) pair.
4. Effect	All FUTURE recommendations for this error type will be auto-executed. Existing recommendations are not retroactively affected — approve them manually if needed.

Auto-Heal Safety

- Auto-heal only runs when a FixPolicyRule exists and has IsAutoHealEligible=true
- The toggle on the Recommendations screen changes the POLICY — it affects all future recommendations for that error type
- The read-only 'Auto-run' badge on each recommendation shows whether auto-heal was enabled at the time that specific recommendation was generated
- Every auto-heal is logged to FixExecutionLog and AuditLog — nothing runs silently
- Recommendations are atomically claimed before execution — concurrent drains cannot double-execute

7 Recommendations & Operator Actions

These are two separate screens with different purposes:

Screen	Purpose
Recommendations (/recommendations)	Your pending action queue. Shows all recommendations with status filters (Pending Review / Approved / Rejected / Executed). This is where you approve, reject, or enable auto-heal. Click any row to open the failure drawer.
Operator Actions (/operator-actions)	Your decision history. One row per Approve/Reject/Retry action, newest first. Shows who made the decision, the outcome, and the current failure status. Useful for shift handoffs and auditing your own activity.

8 Scan Jobs & Manual Triggers

The Scan Jobs screen shows scan run history — one row per completed scan with timing, outcome, and counts.

Action	How
Scan one job	Click the scan icon next to a job, or POST /api/jobscan/{id}
Scan all jobs	Use the Scan All button (triggers all active jobs)
Re-classify	POST /api/jobscan/classify-pending re-classifies unclassified failures and drains pending fixes

9 Quick Reference

Failure Status Meanings:

Status	Meaning
Failed	Newly detected. Awaiting classification and/or operator action.
Resolved	Fixed — either by auto-heal or operator approval. Terminal state.
ManualRequired	Automated fix failed or was rejected. Operator must investigate manually.
AwaitingManualAction	Operator acknowledged a manual recommendation. Off-system work in progress. Use Mark Resolved when done.

Keyboard Shortcuts:

Shortcut	Action
Ctrl/Cmd + K	Open command palette (search screens, failures, jobs)
↑ / ↓	Navigate failure rows (auto-paginates)
Enter	Open failure drawer for focused row
Esc	Close drawer
Tab	Cycle through drawer's action buttons

Top Bar Controls:

Control	Purpose
Worker pill (green/amber)	Shows worker status. Admins can click to pause/resume.
■ Reconnecting chip	Appears when server heartbeat fails. Data shown is stale.
Sun/Moon icon	Toggle dark/light theme
Globe icon	Language switcher (English active; Hebrew coming soon)
Avatar dropdown	Theme, language, sign out

10 Job Flow View (Operator+)

The Flow view is a read-only visualization available on each monitored job in the config screen. It traces every scan rule's path through the system: detect → classify → recommend → fix.

What the Flow view shows:

State	Meaning
Complete path	Green — scan rule has a matching classification rule AND a fix policy. Auto-heal (■) or manual (■) indicated.
Coverage gap	Amber warning — scan rule has no matching classification rule. Dead path — failures will be detected but never classified.
No fix option	Classification exists but no fix policy for that error type. Failures get classified but no recommendation is generated.
Not determinable	For broad rules (ErrorKeyword, FileContent) where the link to a specific classification can't be computed statically. Shown as neutral ■.

The Flow view is a comprehension/audit tool — not an editor. It's useful for verifying that your configuration covers all expected failure paths before going live.

11 Getting Help

If you encounter an issue MAIA can't resolve:

Step	What to Do
Check the error message	The failure drawer shows the raw error message and execution history. Most issues are self-explanatory.
Check execution history	If a fix failed, the ResultDetail field shows WHY — SQL error, HTTP status, 0 rows affected, etc.
Check the Flow view	Verify the job's configuration covers the error path.
Ask your admin	If a failure is unclassified or has no fix policy, an administrator needs to add rules.
Check the Audit Log	Administrators can view the full audit trail at /config/audit.